



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

SEMANTIC WEB · TOPIC 2

Vietnamese DBpedia

Building a DBpedia version
for the Vietnamese language

Group 23

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ONE LOVE. ONE FUTURE.

The Assignment and What We Built

Task	What we built	Result
1. Define an ontology	vio: in OWL, every class under a dbo: class	18 classes, 54 properties
2. Collect articles	entities chosen via Wikidata, articles via the MediaWiki API	990 articles
3. 4-star data	infobox extraction, Unicode IRIs, quality checks, OWL 2 RL	131,233 triples
4. Link to DBpedia	owl:sameAs from the Wikidata sitelink of the same item	777 links
5. SPARQL interface	Linked Data server, SPARQL editor, question answering	4 tabs, HTTP 303

Domain: in depth rather than in breadth

Vietnamese **football** (players, clubs, national teams, stadiums), **provinces** and **universities** — not all of Vietnamese Wikipedia

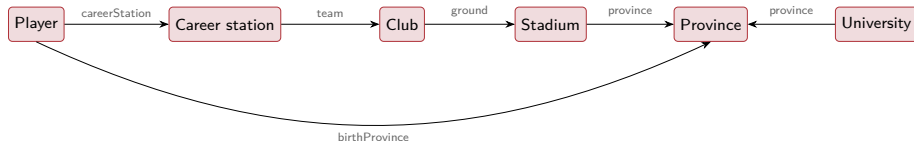
Why This Domain?

990 entities

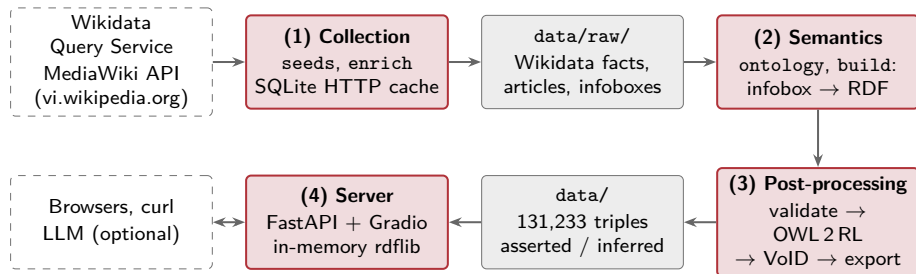
- 611 football players
- 61 clubs, 10 national teams
- 29 stadiums
- 168 universities
- 110 provinces (34 current)
- 1 country

Chosen because they are

- **densely linked** → multi-hop questions
- rich in infoboxes: **97%** of articles
- linkable: **78%** have an English article
- selected through Wikidata (P31, country Q881), not through viwiki categories, which editors assign inconsistently



Architecture: Four Layers, Six Commands



```
python -m vidbpedia <step>
```

```
seeds → enrich → ontology → build → postprocess --strict → serve
```

- Only layer (1) calls the network; raw downloads are committed → the dataset rebuilds **offline**, identically
- Graphs built with the rdflib API; **one function** creates every IRI

Collecting Entities and Articles

Wikidata Query Service — one query per class, e.g. players:

```
?item wdt:P31 wd:Q5 ;                                # human
      wdt:P106 wd:Q937857 ;                             # association football player
      wdt:P27 wd:Q881 .                                  # citizen of Vietnam
?vs schema:about ?item ; schema:name ?viTitle ;
      schema:isPartOf <https://vi.wikipedia.org/> .    # has a Vietnamese article
```

- drop list and disambiguation pages; English article optional
- facts in blocks of 300: dates **with precision**, quantities with point in time, relations with start/end/apps/goals qualifiers
- birthplaces and locations mapped to a province through P131*

MediaWiki API (vi.wikipedia.org), batches of up to 50 titles

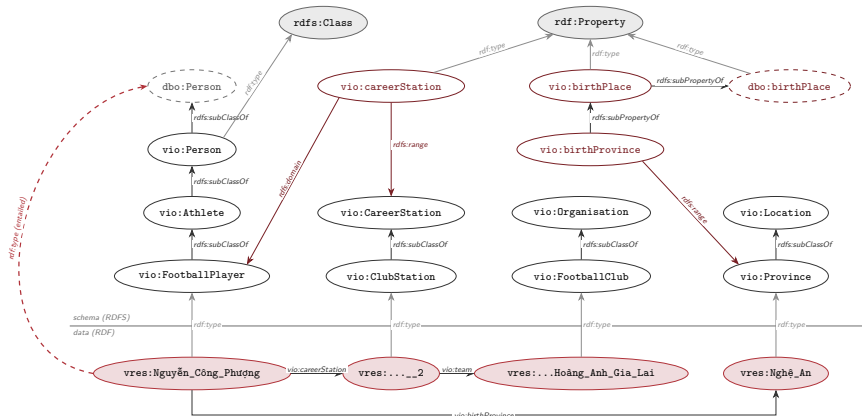
- page ID, revision, image, coordinates, lead section (abstract)
- categories, redirects, infobox parameters, external links
- 2,588 infobox link targets resolved (71 disambiguation pages)

Polite and reproducible

2s / 1s between requests, maxlag, retries on 429 · SQLite cache, --offline and --refresh



Ontology vio: as an RDFS Graph



Excerpt: 18 classes and 54 properties in total, every class $\sqsubseteq$ a `dbo: class` · dashed: `rdfs:type` `dbo:Person` entailed by RDFS

OWL Axioms Do the Work

Axiom	Definition	After reasoning
Property chain	$\text{careerStation} \circ \text{team} \sqsubseteq \text{playedFor}$	2,772 player-team edges
Inverse	$\text{hasPlayer} \equiv \text{playedFor}^{-}$	clubs list their players
$\exists$ restriction	$\exists \text{careerStation.NationalTeamStation} \sqsubseteq \text{NationalTeamPlayer}$	486 national-team players
$\forall$ restriction	$\text{ClubStation} \sqsubseteq \forall \text{team.FootballClub}$	154 more clubs typed
Alignment	every vio: class, 38 properties $\sqsubseteq$ dbo:	dbo:Person: 0 $\rightarrow$ 659
Disjointness	4 owl:AllDisjointClasses	caught 2 modelling bugs

Designed from competency questions

“Which clubs has a player played for?” · “How many provinces are there now?” ·
“Which clubs play at home in Hà Nội?” — each is now an automated test

From Wikitext to RDF

Infobox extraction

- 1 **Render** nested templates (dates, heights, lists); drop refs
- 2 **Parse** Vietnamese values: 16.361,2, 2015-, → (mượn) (on loan)
- 3 **Map** keys to `vio:` for 5 template families
- 4 **Choose** Wikidata or infobox per property

Career rows `yearsn/clubsn/capsn/goalsn`
→ 3,382 CareerStations for 601 players

DBpedia datasets reproduced

- labels (vi, en), abstracts: 990
- page IDs, revisions, provenance
- images: 543
- categories: 9,126 links
- redirects: 1,274
- external links, homepages, coordinates
- raw infobox `vip::` 24,605 triples

IRIs keep Unicode, like DBpedia chapters

`http://vi.dbpedia.org/resource/Nguyễn_Công_Phượng` · stations ...__1,
...__2, ... · redirects → `skos:altLabel`

Transform to 4★: One Infobox, Before and After

Before: wikitext — open and machine-readable, but text that names nothing with a URI ($\leq 3\star$)

```
| birth_date   = {{ngày sinh và tuổi|1995|1|21}}
| birth_place = [[Đô Lương (huyện)|Đô Lương]], [[Nghệ An]], [[Việt Nam]]
| height      = {{convert|1,69|m}}
| years2      = 2016 | clubs2 = → [[Mito HollyHock]] (mượn) | caps2 = 5 | goals2 = 0
```

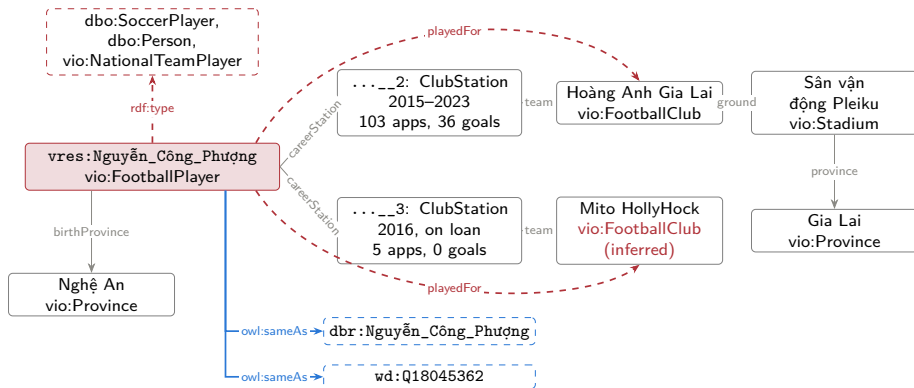
⇓ `python -m vidbpedia build`

After: RDF with an HTTP IRI for every thing (4★)

```
vres:Nguyễn_Công_Phượng a vio:FootballPlayer ;
  vio:birthDate "1995-01-21"^^xsd:date ;
  vio:birthPlace vres:Đô_Lương_(huyện\) ; vio:birthProvince vres:Nghệ_An ;
  vio:height 1.68e+00 ; # xsd:double; Wikidata wins over 1,69
  vio:careerStation vres:Nguyễn_Công_Phượng__3 .
vres:Nguyễn_Công_Phượng__3 a vio:ClubStation ;
  vio:team vres:Mito_HollyHock ; vio:onLoan true ;
  vio:startYear "2016"^^xsd:gYear ; vio:endYear "2016"^^xsd:gYear ;
  vio:appearances "5"^^xsd:nonNegativeInteger ;
  vio:goals "0"^^xsd:nonNegativeInteger .
```

IRIs for the player, places, clubs and the loan spell itself · **typed literals** (`xsd:date`, `gYear`, `nonNegativeInteger`, `boolean`) · links and templates **interpreted** · **shared vocabularies** (`vio: ⊆ dbo:`)

One Entity, End to End



— asserted - - - inferred by OWL 2 RL — link to another dataset
2 of the player's 11 career stations; hasPlayer edges omitted

8 checks before export

TBox (kinds, dbo: ancestors, labels) ·
object vs. datatype values · literal
ranges · required fields · functional
properties · object ranges ·
owl:sameAs targets · plausible values

0 errors, 0 warnings

47 tests pass

What reasoning adds (+41,730 triples,
117 s)

Query	Asserted	Published
?x a dbo:Person	0	659
?x a dbo:SoccerPlayer	0	611
?x a dbo:Place	0	370
?x vio:playedFor ?t	0	2,772
?t vio:hasPlayer ?x	0	2,772
?x dbo:team ?t	0	3,699

Input: types and vio: triples only; no sameAs,
functional or cardinality axioms

Reasoning caught two modelling bugs

- (1) National teams inferred as *clubs*, a disjointness violation → domain of *ground/manager* changed to *Organisation*.
- (2) “Clubs at home in Hà Nội” returned clubs from Đà Nẵng → removed *tenant* $\equiv$ *ground*⁻.

Linking to English DBpedia (5★)

How

- Wikidata item → its English sitelink
→ `owl:sameAs dbr:<title>`
- no matching by label, which is ambiguous for Vietnamese names
- all 990 entities also → `wd:Q...`
- validation: no list/disambiguation targets, no shared targets
- described as two VoID linksets

Class	Linked / total
Players	508 / 611
Clubs, national teams	70 / 71
Stadiums	25 / 29
Universities	89 / 168
Provinces, country	85 / 111
Total	777 / 990

Checked on the live DBpedia

dbr: target	Links	Share
Article	733	94.3%
Redirect	12	1.5%
Not in DBpedia	32	4.1%

Our titles come from today's English Wikipedia; DBpedia is an older snapshot (16 of the 32 missing targets are clubs)

Confirms DBpedia uses the **same Unicode IRIs**, e.g. `dbr:Đặng_Văn_Lâm`

Dereferenceable IRIs: 303 and Content Negotiation

Route	Response
/resource/{name}	303 → HTML page, or → /data/{name}.ext when Accept asks for RDF
/page/{name}	303 → HTML page, like dbpedia.org/page/...
/data/{name}.ext	triples about the resource: .ttl, .nt, .jsonld, .rdf
/ontology/{term}	definition of a vio: term, in Turtle

Recorded on the running server:

```
$ curl -si -H "Accept: text/turtle" \  
    http://127.0.0.1:7860/resource/Nguy%E1%BB%85n_C%C3%B4ng_Ph%C6%B0%E1%BB%A3ng  
HTTP/1.1 303 See Other  
vary: Accept  
location: /data/Nguy%E1%BB%85n_C%C3%B4ng_Ph%C6%B0%E1%BB%A3ng.ttl  
  
$ curl -sL -H "Accept: text/turtle" http://127.0.0.1:7860/resource/Nguy%E1%BB%85n...  
vres:Nguyễn_Công_Phượng a dbo:Animal, dbo:Athlete, dbo:Person, dbo:SoccerPlayer, ... ;  
    dbo:birthDate "1995-01-21"^^xsd:date ;  
    ...  
    vio:birthProvince vres:Nghệ_An ;  
    ...  
    owl:sameAs dbr:Nguyễn_Công_Phượng, wd:Q18045362 ;
```

Headers and Turtle abridged. Browsers get the resource page at `/?resource=<name>`, a shareable URL. Clients must percent-encode the Vietnamese characters, as browsers do.



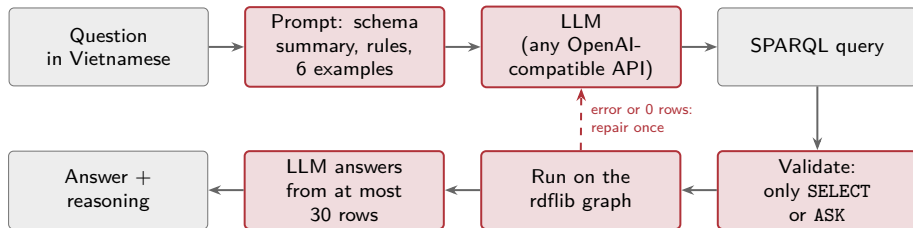
Five Stars, Checked

Star	Evidence
* open licence	CC BY-SA 4.0 in the VoID description and the ontology; dumps in a Git repository
** machine-readable	RDF triples, not text or tables
*** open formats	Turtle and N-Triples dumps; each resource also as JSON-LD and RDF/XML
**** URIs	an HTTP IRI for every entity, station, category and redirect; dereferenced with 303 + content negotiation
***** linked	777 owl:sameAs to DBpedia (733 verified), 990 to Wikidata; classes and properties $\sqsubseteq$ dbo:

Caveat on the fourth star

The `vi.dbpedia.org` namespace follows the DBpedia-chapter convention, but we do not own that domain: the IRIs resolve only through our running server (`http://127.0.0.1:7860/resource/...`).

Question Answering: the LLM Writes SPARQL



- **SPARQL mode:** show the query and the raw rows, skip the answer call
- **Entity lookup by name in a subquery:** rdflib applies FILTER after the join, so the Công Phượng career query drops from **10.7 s** to **0.27 s**
- Works with OpenAI, Gemini or a local Ollama model (.env); tests use a stubbed model, no API key needed

Evaluation: Dataset and Competency Questions

131,233

triples

990

entities

3,382

career stations

0

validation errors

88,898 asserted + 41,730 inferred + 549 ontology + 56 VoID

Question	Answer from the graph	Relies on
How many provinces are there now?	34 (76 former)	FormerProvince
Which unit succeeded Hà Tây?	Hà Nội	successor
Which teams has Công Phượng played for?	11 teams, 7 club stations	property chain
Which provinces have the most players?	Nghệ An 58, Hà Nội 48, Thanh Hóa 42	birthProvince
How many universities are in Hà Nội?	54	province
Who is "Công Phượng"?	Nguyễn Công Phượng	redirect

Every question is an automated test on the real data (lower bounds, so tests survive a re-crawl)

Limitations and Future Work

Limitations

- One domain, 990 entities — each new class needs a seed query, ontology terms, a mapping and a build function
- 44 of 777 DBpedia links are stale (12 redirects, 32 missing)
- 129 career stations without a team IRI; 26 universities without a province
- No SPARQL 1.1 Protocol endpoint (`/sparql`), no query time limit, `SERVICE` not blocked
- Question answering not measured on a question benchmark

Future work

- `/sparql` endpoint with time limits; follow DBpedia redirects at build time
- More classes (coaches, competitions) and page links (`dbo:wikiPageWikiLink`)
- A benchmark of Vietnamese questions for the question-answering tab

Conclusion

A Vietnamese DBpedia for football, provinces and universities

Ontology `vio`: (18 classes, 54 properties) aligned with DBpedia; 990 Vietnamese articles → 131,233 triples, 0 validation errors

Reasoning makes it interoperable

OWL2 RL adds the DBpedia types and properties clients expect, and shortcut edges for multi-hop questions (`playedFor`: 0 → 2,772)

Five-star Linked Data with a query interface

777 links to English DBpedia (733 verified live), 990 to Wikidata; 303 + content negotiation, SPARQL editor, questions in Vietnamese

Code and data: github.com/PhungHaThiKim/vietnamese-dbpedia



Thank you
Questions?